

Yunshu Chen

Medical AI • Clinically-Validated Imaging • Foundation Models • Agentic Decision Support

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PhD candidate in Data Science & AI working on multi-modal medical and biological image analysis across four clinical and industry partnerships. My work asks whether segmentation models are validated at the level that actually changes a decision. For coronary IVUS I moved evaluation from pixel overlap to **clinical measurement agreement**, and am pre-training an IVUS foundation model on **49.6M frames**. Three first-author papers under review.

EDUCATION

Doctor of Philosophy, Data Science & Artificial Intelligence

Oct 2023 – Apr 2027 (expected)

Monash University, Faculty of Information Technology, Melbourne, Australia

- Thesis: multi-modal imaging analysis, vision-language models, and segmentation. Monash Medical AI.

Bachelor of Advanced Computing, Honours Class II Division 1

Feb 2019 – Jun 2023

The University of Sydney, Faculty of Engineering, Sydney, Australia

- Major: Software Development & Computational Data Science. Honours thesis: End-to-Same-End Encryption, a cross-platform blind cloud storage app (React Native, Spring Boot, AWS, Jenkins CI/CD).

PUBLICATIONS & SUBMISSIONS

Y. Chen, et al. "From Pixels to Decisions: Validating Deployable Coronary IVUS Analysis at the Level That Informs Treatment." *First author, under review.*

Y. Chen, et al. "Clinically Aligned Geometry Constraints for Robust IVUS Vessel Boundary Segmentation." *First author, under review.*

Y. Chen, et al. "Confidence-Adaptive Trimap for Boundary-Precise Zebrafish Myotome Segmentation." *First author, under review.*

H. Guo, Y. Chen, X. Xie, G. Xu, W. Guo. "Efficient Conformer-Based CTC Model for Intelligent Cockpit Speech Recognition," 2022. *Published.* doi:10.1109/ISCSLP57327.2022.10037993

RESEARCH EXPERIENCE

Geometry-Consistent Coronary IVUS Segmentation & Clinical Measurement Validation

Jun 2025 – present

Monash University, with the Victorian Heart Institute (CARAT and Huygens cohorts)

- Built an end-to-end pullback analysis system covering QC, side-branch detection, lumen/EEM segmentation and plaque progression, modelling the vessel's annular structure in a **polar-unwrapped domain** with differentiable geometry losses. Dice 0.922–0.930 / HD95 0.275 mm under **5-fold patient-disjoint CV**, statistically indistinguishable from nnU-Net at a fraction of inference cost.
- Agreement with expert measurement: lumen CSA **CCC 0.94**, vessel CSA **CCC 0.96**; plaque-burden treatment threshold **κ 0.67, 84% agreement**; calcium-arc grading ($>180^\circ$, determines rotational atherectomy) **AUROC 0.97–0.99** over 63 patients / 8,717 frames.
- Knowledge distillation produced a model **2.6× smaller with 3× faster CPU inference** and no measurement degradation, trainable on **1/6 of the annotation budget**; uncertainty-guided deferral cut measurement MAE from 0.91 to 0.64. Released a full reproducibility archive (folds, held-out set, expert ground truth, per-frame predictions, reader-study tooling).

IVUS Foundation Model: Multi-Scale Vendor-Locked Supervision

Mar 2026 – present

Monash University. 49.6M frames, 28,753 pullbacks, two vendor platforms, 13 years of clinical acquisition

- Proposed **vendor-locked supervision**: all fine-grained labels originate from the one platform physically capable of producing virtual-histology classes, measurement contours and structured reports, targeting **zero-shot transfer to the ~74%-market-share platform that cannot generate them at all**.
- Four supervision levels from pixel tissue masks (67,203 frames × 4 classes) to patient-level clinical reports (7,267 paired), on a pre-training infrastructure I built end to end: large-scale DICOM→tensor pipeline, **md5-versioned data lineage**, multi-GPU training.

LLM-Orchestrated Clinical Decision Support for IVUS

2026 – present

Monash University. Clinician-reviewed across 35 cases

- Wrapped the foundation model and segmentation stack as **deterministic tools orchestrated by an LLM through a ReAct loop**. Architectural guarantee: **every number in the report comes from an algorithm; none is generated by the language model**.
- JSON-Schema tool registry and Pydantic state ledger make each step auditable; **8 guideline-grounded decision tools** carry ESC/consensus citations. Deployed on a **locally hosted Qwen2.5-7B + vLLM** for patient-data privacy, with bilingual reports and a clinician review interface.

Multi-Scale Cell Instance Segmentation Foundation Model

Feb 2025 – Feb 2026

Monash University, in collaboration with Lead Healthcare

- SAM-based framework with a custom UNETR decoder predicting complementary distance and boundary maps, trained on **100K+ images and 8M+ instances** across four imaging modalities, reaching **89% mAP and outperforming Cellpose-SAM, microSAM, SAMCell and CellSAM across 12 benchmarks**. LoRA fine-tuning adapts to unseen cell types from 5–10 samples; **deployed as a production web application**.

SAM2 Memory as a Tracking Linker

Monash University. Time-lapse microscopy cell tracking

Apr 2026 – present

- Replaced a state-of-the-art tracker's learned association module with **SAM2's memory mechanism: CHOTA 0.8603 vs Trackastra's 0.8511, $p < 0.001$, pure inference with no retraining**; won on 6 of 6 fields of view.

Confidence-Adaptive Trimap for Zebrafish Myotome Segmentation

Monash University, in collaboration with the Australian Regenerative Medicine Institute

Apr 2024 – present

- Derived the uncertain boundary band from the model's own predictive confidence with a per-image adaptive threshold, **replacing hand-tuned morphological kernels**. Dice 0.856 / B-IoU 0.467 / HD95 31.1 on 121 images from 35 fish under **fish-level 5-fold CV**, significantly beating every baseline (paired Wilcoxon). Under domain shift to muscular dystrophy the adaptive variant holds while the fixed-kernel one falls below plain U-Net++. Delivered a PyQt analysis GUI to biologists.

Additional projects

- Multimodal acute ischemic stroke outcome prediction** (Apollo Medical Imaging): 3D ResNet-50 over 7-channel CT perfusion fused with clinical variables via cross-modal attention; treatment-specific heads improved accuracy by 12%.

INDUSTRY EXPERIENCE

Machine Learning Engineer (Part-time)

PredictX Health × Western Health, Melbourne. Chemotherapy toxicity prediction for colorectal cancer

Jan 2026 – present

- Built a single-source-of-truth clinical master table joining an encrypted registry, CT-derived body composition and chemotherapy administration records, covering **302 patients, a 57-column schema and 251 strictly matched registry records**, and derived five clinical outcome variables.
- Established pytest regression coverage and a data dictionary written for non-technical clinical readers; delivered via weekly standups with the clinical partner.
- Feature-selection pipeline (logistic regression, LASSO, Random Forest) identified CT-derived muscle density as the strongest toxicity predictor.

Speech Recognition Algorithm Engineer, Intern (Full-time)

NIO, Intelligent Cockpit & Software Development, Shanghai, China

Jul 2022 – Dec 2022

- Improved multilingual ASR accuracy by **5%** across Chinese and European languages; built automated diagnostics resolving **60+ production recognition failures; integrated models into production vehicle systems**.

Research Intern, Institute of Psychology, Chinese Academy of Sciences, Beijing

Pain Comprehensive Imaging Laboratory. MRI experiment design; automated small-sample MRI pipeline (Python, FSL); few-shot learning under data scarcity.

Mar 2022 – Jul 2022

TEACHING & ACADEMIC SERVICE

Teaching Assistant (Sessional Academic), Monash University

FIT5196 Data Wrangling · FIT5145 Introduction to Data Science

Jul 2025 – present

- Weekly tutorials and labs for **60+ postgraduate students; 92/100 student satisfaction**. Student Representative, Data Science & AI, Faculty of IT (Feb 2026 –).

General Executive, 94th SRC & Student Member of Academic Board, The University of Sydney

2021 – 2022

- Elected with **434 primary votes, among the highest in the SRC's history**. Contributed to the *Sydney in 2032* strategy, the MySydney special admission scheme, and a new academic integrity framework. Formally acknowledged by the Chair of the Academic Board.

AWARDS

- 2nd place**, Track II (Unlimited Model Size, CER 9.86%) and **3rd place**, Track I (Limited Model Size, CER 13.39%), Intelligent Cockpit Speech Recognition Challenge, Singapore 2022.
- Formal acknowledgement from the Chair of the Academic Board, The University of Sydney (2022) · Shanghai Jiao Tong University exchange, GPA 4.0/4.0.

TECHNICAL SKILLS

Deep learning	PyTorch · SAM / SAM2 · DINOv3 · ViT · MAE pre-training · nnU-Net · UNETR · ConvNeXt · LoRA
LLM & agents	ReAct orchestration · vLLM · Qwen2.5 · JSON-Schema tool contracts · local/private deployment
Medical imaging	IVUS / NIRS-IVUS · CT perfusion · MRI / fMRI · DICOM · fluorescence, H&E, EM · radiomics
Clinical validation	ICC · CCC · Bland-Altman · Cohen's κ · AUROC · paired Wilcoxon · patient-disjoint CV
Infrastructure	large-scale DICOM pipelines · data lineage · multi-GPU training · Docker · Kubernetes · AWS
Programming	Python (advanced) · Java · R · SQL · C++ · MATLAB · LaTeX · English, Mandarin
Open source	Genius Labbook , a git-backed research logbook CLI; 135 experiment entries across 6 projects . github.com/CHEN-Yunshu/Genius_Labbook